

Case study:

You are the data science lead for a social media platform called PikPok in which users post short videos to their followers. Your team is responsible for the design of the machine learning models behind the PikPok platform.

- On PikPok, users can scroll on their Home page and watch content posted by the creators they follow.
- Users can also interact with the content through "likes" and comments.
- Moreover, users will also see on their Home pages videos recommended by predictive models (based on their preferences/interactions).
- on PikPok users will post videos along with descriptive *hashtags* (e.g., #food, #funny, #outdoors)
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(a) You are designing the content recommendation system of PikPok. One of the parts of the PikPok's content recommendation system (called PikPokIT) is a Machine Learning model that is continuously updated using millions of pieces of content and activities generated by all users of the platform every day.

Q: Regarding the training process of the PikPokIT models behind its recommendation system, would you use full batch, mini-batch, online learning or a combination of them? Describe and justify your choice.

- For PikPokIT's recommendation system, a blend of mini-batch and online learning is ideal.
- Mini-batch learning efficiently handles large datasets by training on smaller subsets, providing scalability.
- Online learning updates the model with real-time user interactions, keeping recommendations current.
- This combination ensures the system is both adaptable to new trends and scalable for processing vast amounts of data, optimizing performance and relevance in a dynamic content environment.